

PROCESS BOOK

BEYOND THE PITCH

The Data Story of Women's International Football
1956 – 2025

From a 50-year ban to 2 billion viewers — told through 11,177 matches, 10 D3 visualizations, and 70 years of data.

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01

Overview & The Original Idea

Women's football is one of the fastest-growing sports in the world — yet for most of its history it was literally illegal. In 1921, England's FA banned women from playing on affiliated grounds, calling it "**quite unsuitable for females.**" The ban lasted 50 years. For 24 years after the first official World Cup in 1991, players competed for zero prize money. By 2023, 2 billion people watched the Women's World Cup final.

Beyond the Pitch was born from this question: what does 70 years of women's international football data actually show? We built a scrollytelling data visualization website that takes a visitor from the 1921 ban to the 2023 revolution, one chapter at a time.

1.1 Target Audience

Football fans who have heard of women's football but don't know its history. Data-literate readers who appreciate both storytelling and analytical depth. Specifically: people who watched the 2023 World Cup but couldn't name a player from the 1991 edition.

1.2 Core Questions

- How did the global reach of women's football grow from 2 nations (1956) to 238?
- What does the prize money gap look like visualised next to attendance figures?
- Which nations dominate, and has that shifted over seven decades?
- Are there universal patterns in when goals are scored — identical to men's football?
- Who are the greatest individual scorers and what does their legacy data look like?

"The most surprising finding in the data is not the growth — it's the timing. Every major expansion happened within 2–3 years of a World Cup. The World Cup is the engine of the entire sport."

02

Dataset & Data Cleaning Pipeline

We used the **Women's International Football Results** dataset from Kaggle, compiled from official FIFA and confederation records, supplemented with manually compiled World Cup prize money and attendance data from FIFA official communications.

Field	Details	Coverage
Matches	11,177 total records	1956–2025
Teams	238 unique nations/entities	All FIFA members + historical
Goals	2,795 goalscorer records	~75% of matches
Scorers	1,055 unique player names	Where available
Tournaments	47 distinct types	Friendly to World Cup
Regions	6 FIFA confederations	Per official mapping

2.1 Data Cleaning Pipeline

- **Deduplication:** Removed 23 exact duplicate match records (same date, teams, score).
- **Name normalisation:** Standardised 47+ country name variants (e.g. "West Germany" → "Germany", "Chinese Taipei" → "Taiwan"). Extended with 150+ aliases in mapfix.js for the TopoJSON map.
- **Region mapping:** Assigned each of 238 teams to one of 6 FIFA confederations using a manually verified lookup table.
- **Year extraction:** Parsed all date strings (5 different formats found) into consistent YYYY-MM-DD.
- **Coverage flags:** Marked each match year as "has_goal_data" based on whether >40% of that year's matches had goalscorer records.
- **Aggregations:** Pre-computed team stats (wins, draws, losses, goals, win rate, goals per match) and head-to-head matrices for the top 10 nations.
- **Race data:** Generated cumulative win counts per team per year for the bar chart race (15 teams × 70 years).
- **Bundle:** All processed data bundled into a single 64KB bundle.js for zero-server-required deployment.

Known Limitation: Goalscorer data is absent for ~25% of matches, concentrated in 1956–1985 and some African/Asian fixtures. Career totals for some players are therefore underestimates. This is flagged explicitly in the bubble chart tooltip.

03

Exploratory Data Analysis

Before building any visualisation we ran a structured EDA in Python (pandas, matplotlib) to understand distributions, outliers, and what stories the data actually contained versus what we assumed going in.

◆ Nations growth is not linear

We expected steady growth. Instead we found step-changes: 1971 (ban lifted), 1991 (first WC), 2003 (expansion). Each World Cup triggered a burst of new nations.

◆ Goal timing is universal

We tested whether goals clustered at the same minutes as men's football. The answer was yes — 45' and 85'+ spikes are nearly identical. This became a key narrative point in Chapter 8 (Radial Goals Clock).

◆ USA dominance is statistically extreme

465 wins, 78.8% win rate across 590 games. Against Germany: 16–2. Against Japan: 30–2. For context, Brazil men's football has a ~67% win rate over the same period.

◆ Prize money changed suddenly, not gradually

The gap was absolute (100%) for 24 years, then dropped to ~89% in a single cycle (2019→2023). This non-linearity drove the economics dual-bar chart design.

◆ 27 non-FIFA entities in the data

Scotland, England, Wales, Northern Ireland play as separate nations in women's football. West Germany, Yugoslavia, Czechoslovakia, and USSR appear as historical teams. All required special handling in the world map country matching.

04

Design Process & Sketches

4.1 Phase 1 — Dashboard (Rejected)

Initial sketches showed a traditional analytics dashboard: a sidebar with filters, a main panel with 4–5 charts. We rejected this because it treats the user as an analyst rather than a reader. The data has a story — it needed to be told, not explored freely.

4.2 Phase 2 — Scrollytelling Structure (Adopted)

We restructured around a **10-chapter narrative arc**, each answering one question. Modelled on data journalism at NYT Graphics and The Pudding. Scroll triggers chapter reveals; each visualization directly supports its headline claim.

ChQuestion	Visualization Type
01 When was football forbidden?	Annotated area chart
02 Who dominated, in what order?	Animated bar chart race
03 Which nations joined when?	Animated choropleth world map
04 Who beat whom most?	D3 chord / ribbon diagram
05 How did continental power shift?	Streamgraph (6 confederations)
06 What does the pay gap look like?	Dual bar chart (men vs women)
07 Who were the greatest scorers?	Force-directed bubble chart
08 When do goals happen?	Radial clock + coverage bar chart
09 Who dominates by win rate?	Scatter plot with collision labels
10 How do two nations compare?	Head-to-head radar chart

4.3 Phase 3 — Cinematic Sports Editorial Aesthetic

Visual language inspired by Sky Sports broadcast graphics, ESPN design systems, and The Athletic editorial style. We settled on a **dark background (#05050a) with gold (#f5c842) + rose (#ff4d6d) accents** — referencing both the gold medal and the football pitch under floodlights. Typography: Bebas Neue (display) · Oswald (headings) · DM Sans (body).

05

Visualization Design Decisions

◆ Bar Chart Race — Why animate?

A static ranking hides the drama of HOW the USA overtook Germany. Animation makes the viewer feel the 1985 debut and subsequent rise. Speed control lets analysts pause at key moments (1991 WC, 2011 Japan win). 15-team limit avoids visual noise.

◆ Chord Diagram — Why not a heatmap?

A heatmap would show the same data but head-to-head rivalry is fundamentally about connection. The chord makes asymmetry visual — the thick arc from USA to Germany vs the thin arc back is unmissable at a glance. Hover isolation reduces cognitive load.

◆ Streamgraph — Why wiggle offset?

Wiggle offset minimises individual stream slopes, making the overall continental flow more readable. Trade-off: absolute values are harder to read, compensated by hover tooltips showing exact values per year.

◆ Radial Goals Clock — Why polar?

A standard time-series line of goal minutes is familiar but low-impact. The radial form immediately suggests a clock — the correct mental model for match time. The 45' and 85'+ spikes protrude visibly beyond the circle boundary.

◆ Scatter Plot — Why bubbles, not dots?

Encoding goals_for as bubble radius adds a third data dimension. We implemented a 40-iteration collision avoidance algorithm for labels — all 30 nation names are visible with thin leader lines. Standard D3 force layout was too slow.

◆ Force Bubble Chart — Why physics simulation?

Goalscorer data has no natural spatial ordering. The force simulation creates organic, non-overlapping clusters that feel alive. Player name appears on hover to keep the chart legible at all screen sizes.

◆ Hero Before/After Panel — Why not a chart?

The hero section needed to communicate the entire story in 3 seconds. Four paired numbers (BANNED → 2B viewers; \$0 → \$110M; 2 → 238 nations; 90K → 1.98M attendance) achieve this more powerfully than any chart could.

06

Technical Implementation

6.1 Stack

Layer	Technology	Rationale
Data Processing	Python 3.11, pandas 2.0	Fast iteration for EDA; clean JSON output
Visualization	D3.js v7	Full SVG control; required for chord + radial
Map	TopoJSON v3, world-atlas 2	Lightweight; 110m resolution sufficient
Animation	CSS transitions + Canvas API	Canvas for particles; CSS for UI reveals
Scroll Events	Intersection Observer API	Native browser; no GSAP dependency
Data Delivery	Inlined bundle.js (64KB)	Works offline; no CORS/server issues
Fonts	Google Fonts CDN	Bebas Neue, Oswald, DM Sans

6.2 File Structure

- **index.html** — Single HTML file with all 10 chapter sections
- **css/style.css** — 1,200+ line stylesheet with CSS custom properties
- **js/charts.js** — All 11 D3 chart functions (~1,000 lines)
- **js/crossfilter.js** — Cross-chart filter (map click → scatter + race highlight)
- **js/interactions.js** — Scroll reveal, cursor, loader, stat counters
- **js/background.js** — Canvas particle field + pitch geometry
- **js/mapfix.js** — 150+ country name aliases for map matching
- **data/bundle.js** — All datasets inlined (64KB, zero network requests)

6.3 Cross-Filtering System

A lightweight cross-filter module (`crossfilter.js`) connects the world map to both the scatter plot and bar chart race. Clicking a country fires **CF.selectCountry()** which: (1) dims all non-matching scatter dots, (2) dims all non-matching race bars, (3) shows a dismissable gold badge at the top. Implemented without external libraries using D3 selection `.each()` to attach data references to DOM elements.

07

Iterations & Changes

→ **Map Projection: Mercator** → **Natural Earth**

Mercator distorts African and Antarctic regions. Natural Earth better represents the continents where much of the growth story takes place.

→ **Race Chart: 30 Nations** → **Top 15**

Initial version showed all 30+ teams. The chart became unreadable. Filtering to top 15 by total wins preserves all interesting narrative without visual noise.

→ **Hero: Player Cards** → **Before/After Stats** → **Photo Collage**

V1–V7 showed a player photo triptych which duplicated the Legends section. Replaced with Before/After data pairs, then with a 4-photo diagonal collage of current-generation players — completely distinct from the legends podium.

→ **Chatbot: Live API** → **Scripted Story Guide**

API-based chatbot required network/API keys and failed in offline environments. Replaced with 8 pre-scripted story buttons — faster, reliable, pedagogically cleaner.

→ **Epilogue Section** → **Removed**

Credits section was removed to tighten the narrative. The site now ends on the Head-to-Head Explorer, giving users an active final experience.

→ **Scatter Labels: Filtered** → **Full Collision Avoidance**

Only 8 teams were labelled initially. Replaced with a 40-pass iterative collision avoidance algorithm — all 30 nations now labelled with thin leader lines.

→ **Legends Podium: Silhouette SVGs** → **Clean Photo Cards**

Abstract ghost silhouettes overlaying player photos were confusing and visually broken. Removed entirely; cards are now clean photo + goal donut ring + info.

08

Evaluation & Limitations

What Works Well

- The scrollytelling structure keeps users oriented — chapter labels and nav indicator always show position.
- The bar chart race is the most engaging element: in informal testing, viewers spontaneously asked to replay it.
- The economics dual-bar chart makes the pay gap viscerally clear without editorializing — the \$0 labels speak alone.
- The chord diagram handles the complexity of 10×10 rivalry data without overwhelming the viewer.
- All charts render in <200ms on a mid-range laptop; the bundle.js approach avoids network round-trips.
- Cross-filtering between map, scatter, and race creates genuine analytical depth.

Known Limitations

- **Goal data coverage (25% missing):** Some player careers are underestimated. Explicitly flagged in tooltips.
- **Map name matching:** Hong Kong, Chinese Taipei, Kosovo don't perfectly match TopoJSON polygon names.
- **Historical teams:** West Germany, Yugoslavia, USSR appear in data but don't map to current polygons.
- **Responsive design:** The chord diagram and scatter become crowded below 768px. Optimised for 1280px+.
- **Bar race text collision:** At very fast speeds, team name transitions can briefly overlap.

09

Division of Labour & Conclusion

Member	Primary Responsibilities	Charts Owned
Imane Oujja	Data story narrative, history research, process book writing, timeline integration	Ch01, Player Legend · Ch05 Bubble Chart · Hero section
Shyamala Vasireddy	World map implementation, rivalry analysis, cross-filtered system, map classes	Ch03 World Map · Ch04 Chord Diagram · Ch08 Rivalry
Rim Abkari	Data preprocessing pipeline, economics research, technical architecture, CSVs	Ch02 Rivalry Chart · Ch05 Streamgraph · Ch06 Economic

Shared Responsibilities

- D3.js debugging and chart polish — all three members
- Cross-browser testing (Chrome, Firefox, Safari) — all three members
- Content writing for narrative quotes and chapter ledes — collaborative
- Final integration and deployment preparation — Rim Abkari
- Design system (colours, typography, CSS variables) — Imane Oujja
- Data cleaning and Python preprocessing — Shyamala Vasireddy

Conclusion & Future Work

Beyond the Pitch successfully transforms 70 years of raw football data into an accessible, cinematic data story. We proved that data visualization is not just about drawing charts — it is about guiding a user toward a revelation.

If we had an additional semester we would: (1) add sub-national data (club-to-national pipeline analysis), (2) integrate live FIFA API feeds, (3) build a mobile-first responsive layout for the chord and scatter charts, and (4) add audio narration for accessibility.

